

Derek Mackley

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SUMMARY

Staff security engineer who ships code that scales security. Built enterprise vulnerability automation processing 3,800+ containers, RFID access control systems, and security tooling from bare metal to cloud. My GitHub reflects this builder mindset, from Python CLI tools and MCP servers to Kubernetes GitOps clusters. I don't just identify security gaps; I write the code to close them, creating developer-friendly automation that enables rapid, secure deployment across hundreds of applications.

TECHNICAL SKILLS

Security Engineering: Vulnerability scanning automation, SAST/DAST implementation, Container security, Kubernetes security, Secret scanning, SBOM generation, KEV/EPSS enrichment, Threat modeling, Penetration testing

Security Tools: Trivy, SonarQube, Defect Dojo, Burp Suite, Invicti, SOPS, Authentik, Keycloak

Cloud & Infrastructure: AWS, GCP, Azure, Kubernetes, Docker, Helm, ArgoCD, Flux, Terraform, Prometheus, Grafana, Proxmox, Talos

Development & Programming: Python, C#/.NET, JavaScript, PHP, SQL, Angular, AngularJS, React, NestJS, Flask, Laravel, MVC, Webforms, jQuery, Bootstrap, LINQ, Entity Framework, NPM

DevOps & Automation: GitHub Actions, Jenkins, Azure DevOps, Ansible, Git, Linux, Mosquitto MQTT, EMQX, Reloader, Visual Studio, Tera Term

Leadership & Management: Technical team building and hiring, Strategic planning balancing business requirements with technical debt, Software release planning and scheduling, Security-focused SDLC implementation

Certifications: Practical Network Penetration Tester (PNPT), Google Cybersecurity, AINS 21

EXPERIENCE

Mastery Logistics Systems

Remote

Staff Product Security Engineer

July 2023 – Present

- Designed enterprise vulnerability automation platform processing 3,800+ container images across 14 Kubernetes clusters, integrating Trivy, SonarQube, and custom scanners into centralized Defect Dojo with automated Jira ticket generation based on SLA requirements
- Built comprehensive GitHub Actions security pipeline for 400+ repositories with automated scanning for secrets, dependencies, and base images in every PR, preventing security issues before production deployment
- Developed Python CLI tool with multi-threaded async architecture for vulnerability enrichment using EPSS/KEV scores, automated organizational mapping, and cross-platform remediation orchestration
- Architected Kubernetes security monitoring using Trivy Operator, Prometheus, and custom Grafana dashboards, providing real-time visibility of vulnerabilities, misconfigurations, and compliance violations with automated source repository correlation
- Led security incident response including CVE-2023-4911 demonstration, established shift-left security program with developer tooling/documentation, and integrated AI tools (GitHub Copilot, Claude) into AppSec workflows with appropriate security guardrails
- Implemented custom security APIs integrating CISA KEV database and endoflife.date for risk-based vulnerability prioritization, reducing critical vulnerability remediation time by identifying actively exploited vulnerabilities in production environments

Mastery Logistics Systems

Remote

Senior Software Engineering Manager

May 2022 – July 2023

- Worked as the Senior Software Engineering Manager in the International region at Mastery Logistics Systems. In this role I led my teams to successfully complete projects including integrations with ITS, Slack, Teams, an IMDL aggregation tool, and a “flat-file” integration that is being used by multiple domains. While in this role I led as many as 9 teams and have helped to define the SDLC for the international region. This role required the ability to juggle many projects simultaneously as well as facilitate communication between many technical and non-technical team members. While in this role my responsibilities included very hands on technical work to resolve bugs or solve deployment issues, to large scale project planning with multiple teams.

Traction Tools

Remote

Software Engineering Manager

February 2021 – May 2022

- Worked as the Software Engineering Manager for a direct-to-consumer site that provides tooling for businesses running EOS. I led the software, QA, and project management teams, overseeing sprint management, code reviews, tech stack, and the direction of the product from an engineering standpoint. While in this role I instituted Azure Devops, automated deployments, unit testing, integration testing, selenium tests and a two-week sprint/release cadence to the organization. I led the effort to modernize the tech stack, by methodically migrating the website from .NET 4.6.1 MVC Razor and AngularJS, to a microservices architecture with a react front end. This involved moving from AWS virtual machines to an ECS cluster, as well as migrating the current project to .NET Core while ensuring no downtime and adding new features. Basically, rebuilding the car piece by piece while in a drag race.

Satcom Direct (Contract)

Melbourne, FI

Senior Software Engineer

July 2020 – February 2021

- Contract role working with a large array of microservices written in .NET C#. This includes some Azure functions as well as windows forms apps, MVC applications, .NET API's, an Angular 7 portal, and many others. The purpose of all of this is to receive, manipulate, and enhance messages from planes and then send them back to the pilot in real time. Most of the development in this position was “high risk” as any mistakes made could force a plane to land. This creates the need for high coverage of unit and integration tests, with architecture planned around high availability.

Ubicquia

Melbourne, FI

Senior Software Engineer / Development Lead

February 2019 – June 2020

- At this startup, I learned an incredible amount and overcame unique challenges on a day-to-day basis. This role involved creating an enterprise solution to manage cellular connected units on light poles. While in this role, I wrote and managed a PHP Laravel API, with an Angular front end, as well as wrote automation frameworks, MQTT broker connections, python API's, virtualized environments, and a whole host of supporting software. I also learned to manage and configure Linux environments and wrote Ansible scripts to migrate and re-architect the business from Azure to the Google Cloud Platform. The last one saving the company 500K annually. This role involved managing a small team (myself in the back end, 2 people in the FE, and 1 in QA) with big tasks, and short deadlines. This taught me to move quickly, focus on what is truly important, and to write extensible software with manageable tech debt.

Medicomp Inc

Melbourne, FI

Senior Software Developer / Development Lead

May 2018 – February 2019

- While at Medicomp I primarily worked on their direct-to-consumer website AAW that displays reports to doctors nationally. The site was written in webforms and jQuery the challenge being to implement new features and using standard best practices while supporting legacy code. This includes creating new features to use primarily web methods, as well as rewriting pieces of the application in order to facilitate a move to a more modern architecture. I managed a team of four to prioritize work and ensure quality. Before this project I was on a team that attempted to fix an outsourced MVC 5, knockout.js application hosted in Azure that analyzed and diagnosed patient's ECG data.

Starr Companies

Atlanta, GA

Application Developer

August 2017 - May 2018

- Worked on a Document Management/ Reporting web portal. Myself and one other developer rewrote the site in MVC 5 from webforms, moved from an on-prem virtual machine to an azure web service, moved our most resource intensive page to Angular JS, and at the time I left were in the process of updating the front-end to use modern technologies, like Saas, Twitter Bootstrap, NodeJS, etc. In addition, began moving the service layer from .NET4.6.1 to a .NET Core Web API.

Global Associate/IT Associate

March 2016 - August 2017

- As part of the Global Associates program at Starr Companies, I was given the unique opportunity to speak and learn from Industry leaders in order to get a comprehensive knowledge of the Property and Casualty side of the Insurance industry. This included creating multiple POC's most notably a cross selling utility that allowed underwriters to see if other facets of Starr were doing business with the same client. In addition to my studies while in this program, I worked as a developer for the duration of this time on the project listed in the description above, and during this time took the daily user base of 30 underwriters to more than 250 across many facets of Starr.

EDUCATION

Georgia State University, J. Mack Robinson College of Business

Atlanta, GA

Bachelor of Business Administration, Computer Information Systems

January
2015

Minor: Information Security

PERSONAL DEVELOPMENT

- Production HomeLab Infrastructure: 7-node Talos Kubernetes cluster on enterprise hardware (64 cores, 512GB RAM, 4x Tesla P100 GPUs) implementing GitOps with Flux CD, Cilium network security, SOPS/Age encryption, cert-manager, external-secrets with Infisical, and 180TB OpenEBS Mayastor storage with encrypted volsync backups
- Open Source Security Projects: Maintain dapper-cluster GitOps cluster with 30+ production services, automated dependency updates via Renovate, Cloudflare tunnel integration, and reusable Terraform/Ansible security automation frameworks on GitHub
- Hardware Engineering: RFID remote car start prototype, custom mechanical keyboards, four operational 3D printers with Fusion 360 CAD workflows
- Professional Certifications: PADI Divemaster, demonstrating risk assessment and safety protocol expertise applicable to security operations